

Table 1: Complete performance comparison at 100k steps vs SafePO at 5M steps.

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit	
SafePO	1c3s5z	7.64 ± 0.16	9.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	1c3s5z	17.40 ± 0.93	7.15 ± 1.68	0.33 ± 0.27	500k		0
Safe Dreamers	1c3s5z	19.29 ± 0.14	4.23 ± 0.27	0.82 ± 0.02	100k		0
SafePO	1c3s5z	7.79 ± 0.05	9.00 ± 0.00	0.00 ± 0.00	5M		4
MAPPO-Lag	1c3s5z	—	—	—	500k		4
Safe Dreamers	1c3s5z	19.15 ± 0.53	4.83 ± 0.94	0.79 ± 0.12	100k		4
SafePO	2m_vs_1z	2.71 ± 0.18	2.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	2m_vs_1z	19.97 ± 0.02	0.01 ± 0.01	1.00 ± 0.00	500k		0
Safe Dreamers	2m_vs_1z	19.30 ± 0.49	0.14 ± 0.08	0.95 ± 0.04	100k		0
SafePO	2m_vs_1z	3.13 ± 0.01	2.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	2m_vs_1z	19.99 ± 0.00	0.00 ± 0.00	1.00 ± 0.00	500k		1
Safe Dreamers	2m_vs_1z	19.34 ± 0.38	0.14 ± 0.09	0.95 ± 0.03	100k		1
SafePO	2s3z	3.69 ± 0.02	5.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	2s3z	—	—	—	500k		0
Safe Dreamers	2s3z	18.44 ± 0.25	2.28 ± 0.38	0.77 ± 0.04	100k		0
SafePO	2s3z	4.86 ± 0.04	5.00 ± 0.00	0.00 ± 0.00	5M		4
MAPPO-Lag	2s3z	—	—	—	500k		4
Safe Dreamers	2s3z	17.37 ± 1.38	2.98 ± 0.64	0.63 ± 0.15	100k		4
SafePO	2s_vs_1sc	2.44 ± 0.10	2.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	2s_vs_1sc	—	—	—	500k		0
Safe Dreamers	2s_vs_1sc	19.95 ± 0.21	0.23 ± 0.11	0.96 ± 0.03	100k		0
SafePO	2s_vs_1sc	4.21 ± 0.21	2.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	2s_vs_1sc	—	—	—	500k		1
Safe Dreamers	2s_vs_1sc	19.57 ± 0.49	0.31 ± 0.10	0.92 ± 0.06	100k		1
SafePO	3m	2.06 ± 0.10	3.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	3m	—	—	—	500k		0
Safe Dreamers	3m	19.57 ± 0.25	0.35 ± 0.14	0.97 ± 0.02	100k		0
SafePO	3m	2.55 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	3m	—	—	—	500k		1
Safe Dreamers	3m	19.46 ± 0.42	0.60 ± 0.17	0.96 ± 0.03	100k		1
SafePO	3s5z_vs_3s6z	3.47 ± 0.02	8.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	3s5z_vs_3s6z	—	—	—	500k		0
Safe Dreamers	3s5z_vs_3s6z	9.60 ± 0.84	8.00 ± 0.00	0.00 ± 0.00	100k		0
SafePO	3s5z_vs_3s6z	3.86 ± 0.04	8.00 ± 0.00	0.00 ± 0.00	5M		4
MAPPO-Lag	3s5z_vs_3s6z	—	—	—	500k		4
Safe Dreamers	3s5z_vs_3s6z	13.28 ± 1.10	7.54 ± 0.40	0.11 ± 0.06	100k		4
SafePO	3s_vs_3z	6.04 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	3s_vs_3z	—	—	—	500k		0
Safe Dreamers	3s_vs_3z	19.83 ± 0.20	0.95 ± 0.18	0.96 ± 0.02	100k		0
SafePO	3s_vs_3z	6.28 ± 0.10	3.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	3s_vs_3z	—	—	—	500k		1
Safe Dreamers	3s_vs_3z	19.93 ± 0.20	0.79 ± 0.48	0.96 ± 0.04	100k		1
SafePO	3s_vs_4z	4.39 ± 0.03	3.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	3s_vs_4z	—	—	—	500k		0
Safe Dreamers	3s_vs_4z	18.19 ± 1.57	2.64 ± 0.34	0.23 ± 0.16	100k		0
SafePO	3s_vs_4z	4.55 ± 0.01	3.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	3s_vs_4z	—	—	—	500k		1

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Table 1 – continued

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit	
Safe Dreamers	3s_vs_4z	16.96 $\pm$ 0.74	2.47 $\pm$ 0.35	0.31 $\pm$ 0.11	100k		1
SafePO	3s_vs_5z	3.44 $\pm$ 0.03	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M		0
MAPPO-Lag	3s_vs_5z	—	—	—	500k		0
Safe Dreamers	3s_vs_5z	16.88 $\pm$ 1.58	2.67 $\pm$ 0.40	0.15 $\pm$ 0.19	100k		0
SafePO	3s_vs_5z	3.52 $\pm$ 0.02	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M		1
MAPPO-Lag	3s_vs_5z	—	—	—	500k		1
Safe Dreamers	3s_vs_5z	17.44 $\pm$ 1.49	2.67 $\pm$ 0.43	0.17 $\pm$ 0.22	100k		1
SafePO	8m	1.43 $\pm$ 0.01	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M		0
MAPPO-Lag	8m	8.94 $\pm$ 0.01	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k		0
Safe Dreamers	8m	18.64 $\pm$ 2.02	2.81 $\pm$ 1.91	0.86 $\pm$ 0.21	100k		0
SafePO	8m	1.63 $\pm$ 0.01	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M		4
MAPPO-Lag	8m	8.91 $\pm$ 0.37	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k		4
Safe Dreamers	8m	17.26 $\pm$ 1.04	5.03 $\pm$ 0.42	0.69 $\pm$ 0.10	100k		4
SafePO	MMM	1.87 $\pm$ 0.01	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M		0
MAPPO-Lag	MMM	—	—	—	500k		0
Safe Dreamers	MMM	17.05 $\pm$ 2.36	6.77 $\pm$ 1.79	0.59 $\pm$ 0.25	100k		0
SafePO	MMM	1.89 $\pm$ 0.04	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M		4
MAPPO-Lag	MMM	—	—	—	500k		4
Safe Dreamers	MMM	17.02 $\pm$ 1.32	7.88 $\pm$ 0.16	0.55 $\pm$ 0.07	100k		4
SafePO	bane_vs_bane	19.26 $\pm$ 0.01	22.55 $\pm$ 0.02	0.79 $\pm$ 0.01	5M		0
MAPPO-Lag	bane_vs_bane	—	—	—	500k		0
Safe Dreamers	bane_vs_bane	17.75 $\pm$ 2.82	13.91 $\pm$ 7.11	0.62 $\pm$ 0.39	100k		0
SafePO	bane_vs_bane	19.37 $\pm$ 0.09	22.40 $\pm$ 0.06	0.80 $\pm$ 0.02	5M		4
MAPPO-Lag	bane_vs_bane	—	—	—	500k		4
Safe Dreamers	bane_vs_bane	19.91 $\pm$ 0.11	6.37 $\pm$ 1.20	0.98 $\pm$ 0.02	100k		4